

Mahmoud Sameh

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- AI Researcher and Electrical Engineer, graduated **top of class** with a **4.95/5 GPA** from Islamic University of Madinah; **IELTS Band 8.0**.
- Research in foundation model adaptation, computer vision, and applied ML, with **three published papers** and **one under review**.
- Selected among the **top 100 nationally** for the KAUST AI Summer Training Program.
- Co-winner of **SDAIA's Enjaz Hackathon** (150K SAR prize).
- Awarded **2nd Best Graduation Project** in AI university-wide.

EDUCATION

Bachelor of Science (B.S.) in Electrical Engineering

August 2020 – June 2025

Islamic University of Madinah, Madinah, Saudi Arabia

- GPA 4.95/5 ([Academic Record](#)) ([External Courses](#))
- Accredited Engineer, Saudi Council of Engineers (SCE) – Membership No. 1213941

LANGUAGE PROFICIENCY

IELTS Academic: Overall Band **8.0** (L: 8.5, R: 8.5, W: 7.0, S: 7.0) ([Certificate](#))

Duolingo (DET): Score **150/160** ([Certificate](#))

WORK EXPERIENCE

AI Engineer (Full-time Internship)

October 2025 – February 2026

Aajil Fintech

- Built AI infrastructure to automate credit risk assessment, reducing manual review overhead.
- Developed a feature-engineering pipeline that generates financial analyses from raw application data.
- Trained and deployed classification/extraction models with end-to-end MLOps.
- Bridged SMEs and the AI team, translating domain feedback into actionable model improvements.

AI Co-Innovation Program

September 2025

Aramco & RDIA

- Participated in a month-long applied AI program focused on **methane and gas detection** from remote sensors.
- Investigated the use of **hyperspectral imaging** and deep learning for spectral-signature identification of fugitive emissions, building on prior HSI hardware/software experience at KFUPM.

Research Intern (Full-time Internship)

June 2025 – August 2025

King Fahd University of Petroleum and Minerals

- Designed a complete experimental protocol for a novel hyperspectral imaging dataset of date fruits, covering sample scanning, oven-drying ground truth, and data management.
- Developed the full software stack for the lab's push-broom HSI hardware, integrating camera control, stepper motor synchronization, and data acquisition into a single Python application.
- Created an open-source GUI (PyQt6) for hyperspectral data calibration, cube assembly, and analysis, published as a first-author paper in *SoftwareX* (doi: [10.1016/j.softx.2026.102680](https://doi.org/10.1016/j.softx.2026.102680)).

- Managed the Engineering College's social media, producing content that increased student engagement.
- Led student teams on college-level technical projects, offering mentorship and coordination.
- Provided student-representative feedback to faculty on program initiatives and curriculum.

PUBLICATIONS

- [1] A. M. Alenezi, **M. Sameh**, M. Aljohani, and H. S. Alharbi, "Predicting emergency department patient arrivals at hospitals using machine learning techniques," *MDPI Healthcare*, 2026, doi: [10.3390/healthcare14091191](https://doi.org/10.3390/healthcare14091191).
- [2] A. BenAbdenour, **M. Sameh**, B. A. Khawaja, A. K. Vallappil, A. M. Alenezi, Q. H. Abbasi, and S. Qazi, "Real-time human fall detection from video using YOLOv11 with pose estimation: A paradigm shift towards efficient transformer-based architectures," *IEEE Access*, 2026, doi: [10.1109/ACCESS.2026.3674843](https://doi.org/10.1109/ACCESS.2026.3674843).
- [3] **M. Sameh**, A. BenAbdenour, and J. K. Ali, "Efficiently adapting SAM 2 for automated schematic capture from hand-drawn circuit diagrams," *IEEE Trans. on CAD of Integrated Circuits and Systems*, no. TCAD-2025-0847, revision under preparation. [[Abstract](#)]
- [4] **M. Sameh**, A. Albeladi, and A. Fawzy, "HSI Control Suite: An integrated GUI for operating and acquiring data from DIY push-broom hyperspectral imaging systems," *SoftwareX*, 2026, doi: [10.1016/j.softx.2026.102680](https://doi.org/10.1016/j.softx.2026.102680).

SKILLS

AI/ML: Deep Learning, Computer Vision, Document Intelligence, NLP/LLM Integration, Data Preprocessing

Languages: Python (PyTorch, TensorFlow, Pandas, OpenCV), C, SQL, JavaScript

Engineering: Control Theory, Digital Signal Processing, Embedded Systems, Robotics

Tools: MATLAB/Simulink/Simscape, Git, Django, HTML/CSS

Project Portfolio: [LinkedIn Projects](#) | [GitHub Repositories](#)

COURSES

- [Harvard CS50: AI with Python](#), [Intro to CS](#), [Understanding Technology](#)
- KAUST Academy: [Intro to AI](#) & [Advanced AI](#) (Passed All 3 Stages)
- [KFUPM: Research Skills](#) (10-Hour Course)
- [Deep Learning Specialization](#) (5 Courses, deeplearning.ai)
- [Michigan Python Specialization](#) (5 Courses)
- [Mathematics for ML & Data Science Specialization](#) (2 Courses)
- [Essentials of Survey Methodology in Scientific Research](#)
- [Entrepreneurship Program](#) (36-Hour Intensive)

Full list with certificates: [Direct Link](#)

PARTICIPATION & ACKNOWLEDGMENT

- Aramco & RDIA: AI Co-Innovation (September 2025) ([Certificate](#))
- Acknowledgment of Academic Excellence (May 2025) ([Certificate](#))
- Letter of Appreciation, Control Lab Redesign (April 2025) ([Letter](#))
- 13th International Cultural Festival (April 2025) ([Certificate](#))
- KSU Engineering Day Competitions (February 2025) ([Certificate](#))
- Enjaz Hackathon (January 2025) ([Certificate 1](#)) ([Certificate 2](#))
- WRO Saudi 2023 RoboSport (October 2023) ([Certificate](#))
- International Conference on Cyber Terrorism (December 2022) ([Certificate](#))